

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-4095
Project Name	Student Enrollment & Course Management System (SECMS)
Maximum Marks	2 Marks

Technology Stack Details

SECMS is built entirely on the Salesforce platform. The table lists the technology chosen for each layer and the purpose it serves in the project.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	User Interface (Frontend)	Salesforce Lightning	Provides the app, tabs, forms and record pages through which users work with student, course, instructor and enrollment records.
2	Logic / Automation (Backend)	Salesforce Flows	Automates enrollment date setting, notifications, instructor assignment and follow-up tasks.
3	Database / Data Storage	Salesforce Objects	Custom objects (Student, Course, Instructor, Enrollment) with lookup relationships store the academic data.
4	Data Validation	Validation Rules	Protect data integrity, e.g. student email condition and fees-paid check before approval.
5	Approval Automation	Approval Process	Routes requested enrollments for controlled approval or rejection.
6	Reporting	Salesforce Reports	Operational reports on student status, enrollments by course, pending approvals and revenue.
7	Visualization	Salesforce Dashboards	Student Management Dashboard combining charts, gauge and table.
8	Security	Profiles, Permission Sets, Sharing / OWD	Role-based access to records and functions for administrators, enrollment officers and instructors.
9	Cloud / Hosting	Salesforce Developer Edition org	Cloud workspace in which the whole application is configured and tested.